Activity No. 2: Interactive Widget	
Course Code: CPE 031	**Program:** Computer Engineering
Course Title: Visualization and Data Analysis	**Date Performed:** July 27, 2026
Section: CPE21S1	**Date Submitted:** July 27, 2026
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1. Discussion

The activity focused on learning how data is processed and transformed before it can be used for analysis. It showed the different steps involved in handling data, starting from collecting raw data, choosing the data format such as **XML, CSV, or JSON**, cleaning the data, and correcting errors in text and numbers.

2. Materials and Equipment

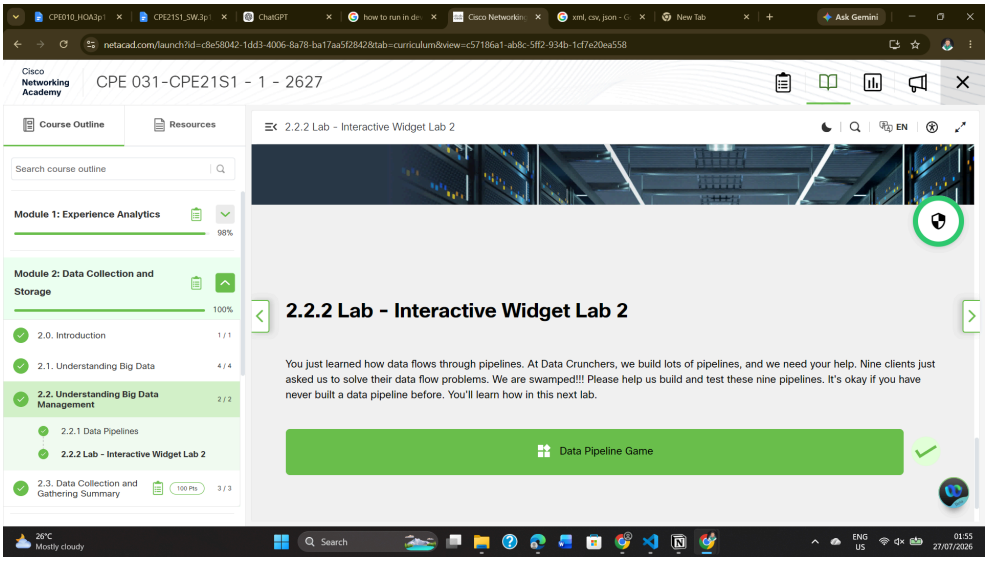
The materials and equipment used in this activity were:

- **Laptop** – used to access the NetAcad platform and complete the interactive activity.
- **Internet Connection** – required to access the Cisco Networking Academy website and load the module.
- **Web Browser** – used to open and navigate through the Introduction to Data Science course.

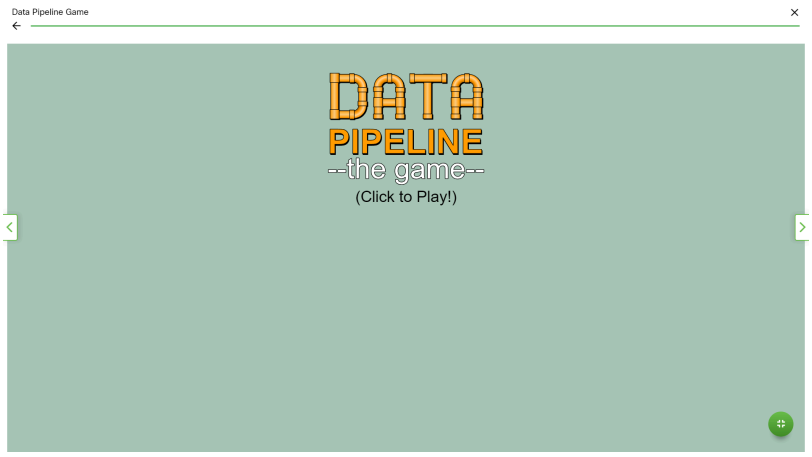
3. Procedure

The procedures performed in the activity were:

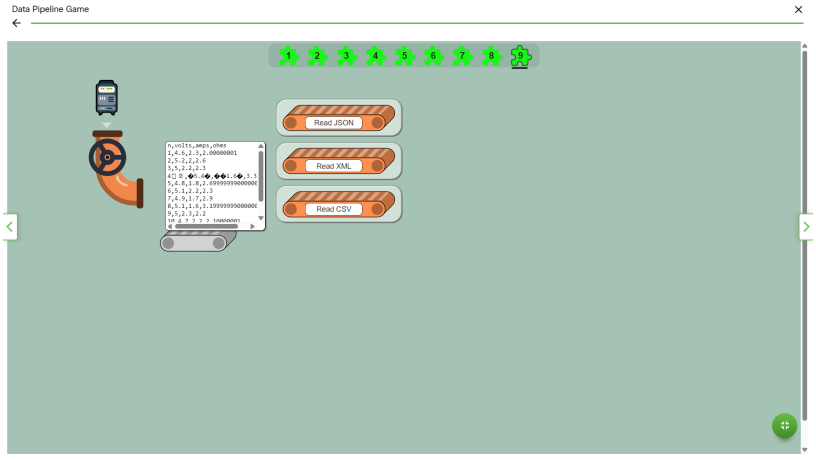
1. I accessed the Introduction to Data Science Course in Cisco Networking Academy and opened Module 2.2.2 Lab – Interactive Widget.



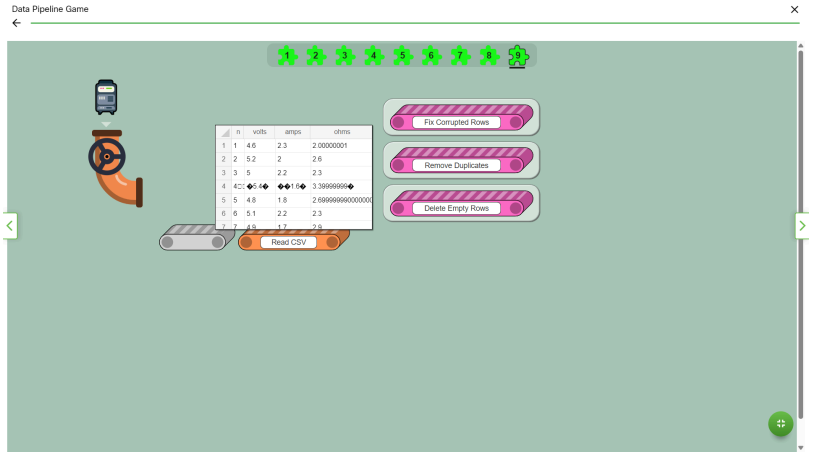
2. I click the start button to proceed to the activity.



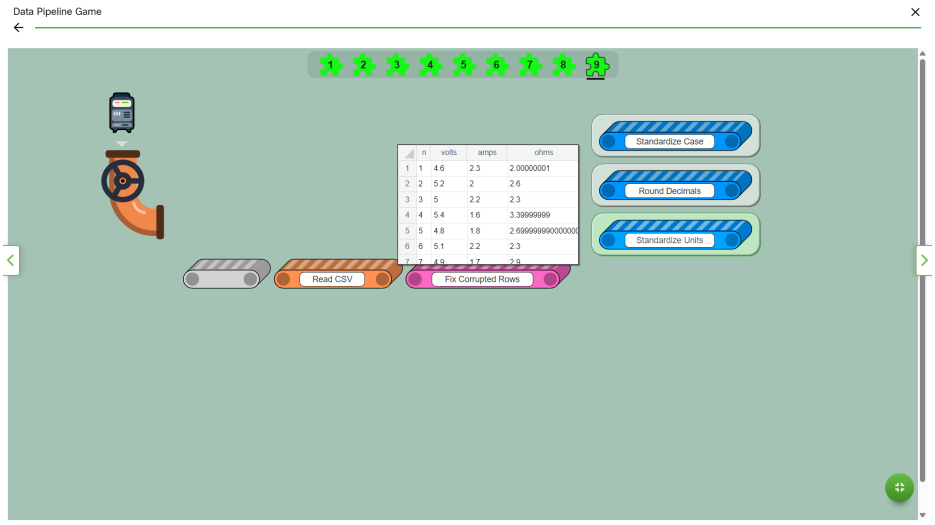
3. I selected the type of data format to be used, including **XML, CSV, or JSON**.



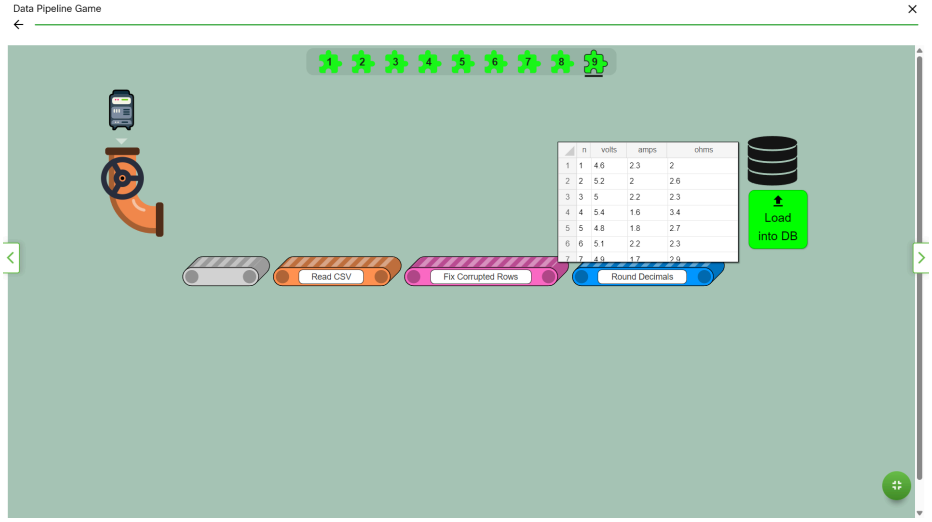
4. I chose the appropriate data cleaning process needed for the dataset.



5. I modified and corrected the data by revising incorrect text entries and adjusting numerical values

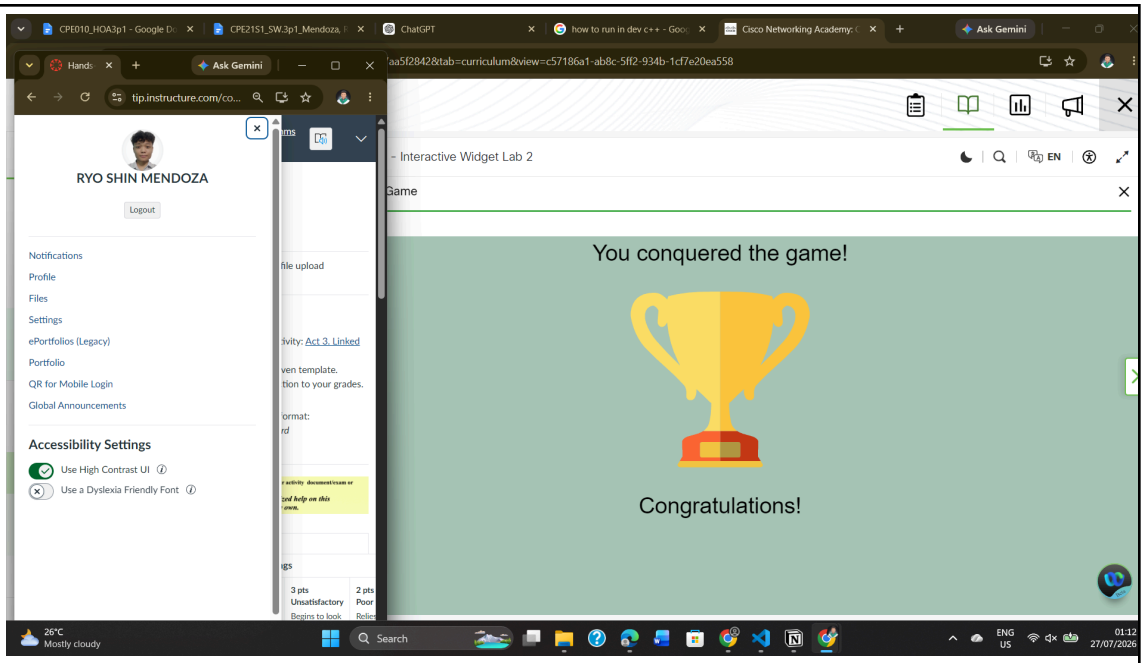


6. I uploaded the processed data into the database provided in the activity.



7. I observed the animation showing how the raw data was transformed into a structured and usable form.

4. Output



5. Conclusion/Analysis/Learnings

Through the interactive widget, I learned that data needs to be organized and cleaned first to make it accurate and useful. After the cleaning process, the data can be stored in a database and transformed into a better format. This activity helped me understand how data scientists prepare raw data before using it for analysis and decision-making.

6. Assessment Rubric

Lab Activity Rubric

Criteria	Ratings						Pts
SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6 pts
SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts
SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24